IEG co., Ltd.

Semiconductor Front-End 6D Digital Twin Simulator

Instructor Script Guide

10 Sessions · 90–150 min each · IEG Training Series

IEG Co., Ltd. — Global Business Division

I. Program Overview

This guide provides session-by-session scripts for teaching the eight front-end stages (Wafer Fabrication → Oxidation → Photolithography → Etching → Deposition → Ion Implantation → Metallization → EDS) using IEG's 6D Digital Twin Simulator.

Defining the '6D Digital Twin'

- 3D — a realistic spatial model of actual equipment and wafer geometry
- 4D — reflects process progression along a time axis (temperature/pressure change, wafer state transitions)
- 5D — a data/intelligence layer where AI predicts and warns of anomalies from accumulated process data (PROCESS ANOMALY DETECTED alerts, etc.)
- 6D — a spatial pattern intelligence layer that reads the EDS wafer map's spatial defect pattern (Edge · Center · Donut · Scratch · Random · Local Cluster) to narrow down root-cause candidates

Session Overview

Session	Topic	Key Simulator Module	Duration
0	Orientation · 6D Digital Twin Overview	Full Dashboard / 3D View Navigation	90 min
1	Wafer Fabrication (Ingot Growth & Slicing)	STEP 01 · Ingot Growth & Slicing	120 min
2	Thermal Oxidation	STEP 02 · Oxide Layer / Thickness Sensor	120 min
3	Photolithography	STEP 03 · Mask Align / Overlay Sensor	120 min
4	Etching	STEP 04 · CD Measurement / Etch Profile	120 min
5	Deposition (CVD/PVD/ALD)	STEP 05 · Thin-Film Uniformity	120 min
6	Ion Implantation	STEP 06 · Dose/Energy Control	120 min
7	Metallization	STEP 07 · Via/Line Resistance	120 min
8	EDS (Electrical Die Sorting)	STEP 08 · Wafer Map / Yield Sort	120 min
9	Integration · 8-Process Chain Scenario & Certification	Full Digital Twin Dashboard (fully linked)	150 min

** Each session runs in three stages: a 10–15 minute intro, 60–100 minutes of development/simulator practice, and 15–20 minutes of wrap-up and checkpoint review.*

II. [Session 0] Orientation · 6D Digital Twin Overview

Learning Objectives

- Understand the overall flow of the eight front-end process stages.
- Explain the '6D digital twin' concept (3D geometry + 4D time axis + 5D data/AI prediction + 6D spatial pattern intelligence).
- Read each of the simulator dashboard indicators (TEMPERATURE · PRESSURE · YIELD · DEFECT RATE · YIELD TREND).

Intro

> **Instructor:** *From today, we experience the entire journey of a single wafer through eight process stages, ending as a finished chip — without ever setting foot in a real fab.*

> **Instructor:** *Let's look at the screen together. This 3D model captures not just the physical geometry of real equipment, but also '4D' data that changes over time, a '5D' layer where AI predicts anomalies, and, later in this program, a '6D' layer that reads spatial patterns on the wafer map.*

Development — Dashboard Reading Training

Point to each of the five indicators on the right-hand panel (Temperature, Pressure, Yield, Defect Rate, Yield Trend) one at a time, explaining which kind of process anomaly each one connects to.

> **Instructor:** *TEMPERATURE and PRESSURE here are physical quantities monitored in common across most processes. YIELD and DEFECT RATE, on the other hand, are indicators that appear as a result after the process finishes. Distinguishing between these two is today's core skill.*

Exercise: learners look at arbitrary values on the dashboard and complete a worksheet classifying each as an 'in-process anomaly' or a 'process-result anomaly.' (5 questions, 10 min)

Mapping the Full 8-Process Flow

On a whiteboard or tablet, list the eight processes in order and have learners summarize in one line what physical/chemical change happens to the wafer at each stage. This map is built up and refined continuously across Sessions 1–8.

Wrap-Up

> **Instructor:** *The 8-process map you drew today is your navigation for the next eight weeks. Starting next session, we'll operate the simulator hands-on, beginning with the first stage: Wafer Fabrication.*

Checkpoint Q&A

Q. What are the two elements the '5D' in a 5D digital twin refers to?

A. Beyond 4D (process change over time), 5D refers to a data/AI-driven layer for predictive maintenance and anomaly prediction.

Q. What is the difference between YIELD and DEFECT RATE?

A. YIELD is the proportion of chips functioning normally (a result), while DEFECT RATE is the incidence of defects found during the process (a root-cause tracing indicator).

II-1. [Session 1] Wafer Fabrication (Ingot Growth & Slicing)

Learning Objectives

- Explain the physical/chemical principles of the Wafer process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final wafer map — the 6D fault case.

Intro

> **Instructor:** *Today covers the starting point of every semiconductor: how a wafer is made from a silicon ingot. We'll start with the physical principles behind turning high-purity silicon into a perfect disc.*

Development — Simulator-Linked Instruction

- Demonstrate the Czochralski growth method in 3D view — play the STEP 01 sequence where a seed crystal rotates and is pulled from molten silicon as a single-crystal ingot grows.
- Show, in slow motion, how a diamond wire saw slices the ingot into thin wafers, and explain how slicing-thickness error affects downstream processes.
- Learner exercise: adjust the simulator's slicing-thickness slider and observe how wafer warpage is reflected in the TEMPERATURE and DEFECT RATE indicators, recording results on the worksheet.

⚠️ PROCESS ANOMALY SCENARIO — Ingot Dislocation Causing Wafer Warpage

Root cause: An imbalance between seed rotation speed and pulling speed introduces dislocations into the crystal lattice.

Symptom (dashboard reaction): DEFECT RATE spikes sharply; the 3D view highlights fine surface curvature on the wafer.

Instructor action: Have learners readjust the rotation-speed parameter and demonstrate the recovery to normal range themselves.

◆ 6D FAULT CASE — Expected Pattern: Center

Dislocation driven by a rotation/pulling-speed imbalance originates at the crystal's rotation axis, so the resulting strain concentrates near the center of each wafer sliced from that section of the ingot.

> **Instructor:** *If this imbalance persisted through several ingot sections, what part of the wafer map would you expect the defect cluster to sit in — near the center, or spread evenly across the whole map?*

Wrap-Up

> **Instructor:** *Today's Wafer process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 9's integrated exercise.*

Checkpoint Q&A

Q. What two process variables determine crystal quality in the Czochralski method?

A. Seed crystal rotation speed and pulling speed.

Q. What is a typical downstream effect of wafer thickness error?

A. Reduced surface flatness can cause film non-uniformity and exposure focus defects in later processes.

II-2. [Session 2] Thermal Oxidation

Learning Objectives

- Explain the physical/chemical principles of the Thermal process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final wafer map — the 6D fault case.

Intro

> **Instructor:** *Today covers the process that forms SiO_2, the oxide layer that protects and insulates the wafer surface. This single thin film determines a device's insulation performance.*

Development — Simulator-Linked Instruction

- Write the dry and wet oxidation chemical reactions on the board, explaining the difference in reaction rate.
- On the STEP 02 screen, play the furnace interior temperature rising through 800–1200°C, comparing the TEMPERATURE graph side by side with the oxide-thickness growth curve.
- Learner exercise: run a mini game-style exercise adjusting oxygen partial pressure and temperature to find the optimal condition for reaching a target thickness (e.g., 100 Å).

⚠️ PROCESS ANOMALY SCENARIO — Oxide Non-Uniformity Causing Gate Leakage

Root cause: Uneven zone-to-zone temperature distribution inside the furnace causes the oxidation rate to vary by wafer position.

Symptom (dashboard reaction): DEFECT RATE rises; the Yield Trend graph turns downward.

Instructor action: Have learners enter zone-by-zone correction values themselves to restore uniformity.

◆ 6D FAULT CASE — Expected Pattern: Edge

Furnace zone non-uniformity is a classic edge effect — heat loss and gas flow differ at the boundary of the process zone, so wafers positioned at the edge of the affected zone show the deviation.

> **Instructor:** *This is the first of four modules in this program that can produce an Edge-shaped pattern on the final wafer map. Given only the shape, could you tell whether the cause was here at Oxidation, or somewhere else? What would you need to check to be sure?*

Key teaching point: Oxidation, Deposition, Etching, and Metallization can all produce an Edge pattern for their own distinct physical reasons. The shape narrows the search to these four candidates — it does not, by itself, name the station. Confirming the cause always requires that station's own dashboard data.

Wrap-Up

> **Instructor:** *Today's Thermal process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 9's integrated exercise.*

Checkpoint Q&A

Q. What is the difference between dry and wet oxidation?

A. Dry oxidation uses only O_2, growing a dense film slowly; wet oxidation uses steam (H_2O) for a faster growth rate.

Q. What is a typical device failure from oxide non-uniformity?

A. Increased gate leakage current and reduced dielectric breakdown voltage.

II-3. [Session 3] Photolithography

Learning Objectives

- Explain the physical/chemical principles of the Photolithography process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final wafer map — the 6D fault case.

Intro

> **Instructor:** *Today's process transfers the circuit design onto the wafer: photolithography. You'll get a feel for why nanometer-scale alignment precision matters.*

Development — Simulator-Linked Instruction

- Play the three-stage sequence — photoresist (PR) coating → exposure → development — on the STEP 03 screen.
- Artificially introduce an overlay error between the photomask and the wafer in the simulator, and have learners observe in 3D how the circuit pattern shifts out of place.
- Learner exercise: adjust the overlay correction parameters (X/Y offset, rotation angle) to bring the alignment error within tolerance (e.g., ±5 nm).

⚠️ PROCESS ANOMALY SCENARIO — Overlay Drift Causing Pattern Misalignment

Root cause: Thermal expansion of the wafer stage, or wafer warpage carried over from an earlier process, shifts the alignment reference point.

Symptom (dashboard reaction): In 3D view, the circuit pattern appears shifted relative to the layer beneath it, and a DEFECT RATE warning appears.

Instructor action: Have learners reset the alignment-correction algorithm parameters themselves and practice recovering from the error.

◆ 6D FAULT CASE — Expected Pattern: Donut

Overlay and focus errors often vary by exposure field/radial zone rather than uniformly across the whole wafer — a specific mid-radius zone can be systematically worse while the center and edge stay in spec.

> **Instructor:** *If one particular exposure zone consistently drifts while zones closer to the center and the outer edge stay within tolerance, what shape would that leave on the wafer map?*

Photolithography and Ion Implantation share this Donut signature for similar reasons — both involve a scan/exposure field that can behave differently by radial zone.

Wrap-Up

> **Instructor:** *Today's Photolithography process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 9's integrated exercise.*

Checkpoint Q&A

Q. What are the three core steps of photolithography?

A. Photoresist coating, exposure, and development.

Q. What are common causes of overlay error?

A. Wafer stage thermal expansion, or wafer deformation carried over from an earlier process.

II-4. [Session 4] Etching

Learning Objectives

- Explain the physical/chemical principles of the Etching process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final wafer map — the 6D fault case.

Intro

> **Instructor:** *Today's process removes unwanted material according to the exposed pattern: etching. The difference between dry and wet etching, and CD (Critical Dimension) control, are central.*

Development — Simulator-Linked Instruction

- Compare wet etching (chemical solution) and dry etching (plasma) directionality — isotropic vs. anisotropic — using the STEP 04 cross-section view.
- Reproduce over-etch and under-etch conditions separately in the simulator, and have learners observe how the CD (Critical Dimension) measurement deviates.
- Learner exercise: adjust plasma power, gas flow, and etch time to derive the optimal recipe for reaching the target CD.

⚠️ PROCESS ANOMALY SCENARIO — CD Deviation from Over-Etch / Under-Etch

Root cause: Excess etch time (over) or insufficient gas flow (under) pushes the pattern linewidth outside the design target.

Symptom (dashboard reaction): The CD measurement graph exceeds the upper or lower spec limit; DEFECT RATE rises.

Instructor action: Have learners readjust the etch-time and gas-flow parameters themselves to bring CD back within spec.

◆ 6D FAULT CASE — Expected Pattern: Edge

Plasma density in an etch chamber is rarely perfectly uniform across the wafer — a well-documented center-fast/edge-slow (or reverse) etch-rate gradient means edge dies are frequently the first to show CD deviation.

> **Instructor:** *This is the second module in this program that can produce an Edge-shaped pattern — but for a different physical reason than Oxidation's zone non-uniformity. How would you use this station's own dashboard data to confirm it's Etching and not one of the other three Edge-capable modules?*

Wrap-Up

> **Instructor:** *Today's Etching process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 9's integrated exercise.*

Checkpoint Q&A

Q. Why is dry etching preferred for fine-pattern formation over wet etching?

A. Anisotropic etching produces vertical sidewall profiles with minimal undercut.

Q. What is Critical Dimension (CD)?

A. The critical linewidth of the circuit pattern formed after etching — a key indicator governing device performance.

II-5. [Session 5] Deposition (CVD/PVD/ALD)

Learning Objectives

- Explain the physical/chemical principles of the Deposition process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final wafer map — the 6D fault case.

Intro

> **Instructor:** *Today's process deposits an atomic-scale thin film on the wafer surface: deposition. You'll get hands-on experience with the differences between CVD, PVD, and ALD.*

Development — Simulator-Linked Instruction

- Play the CVD (Chemical Vapor Deposition) sequence — from reactive gas injection to film formation — on the STEP 05 screen, and compare it against PVD (sputtering) deposition rate and step coverage in a summary table.
- Demonstrate ALD's self-limiting reaction principle with an atomic-scale animation.
- Learner exercise: hold a mini debate selecting which of the three deposition methods best suits a specific structure (a high-aspect-ratio trench), with reasoning.

⚠️ PROCESS ANOMALY SCENARIO — Poor Step Coverage from Film Non-Uniformity

Root cause: Uneven gas flow, or wafer-surface temperature variation, leaves the film thinner than target in certain regions.

Symptom (dashboard reaction): The Thin-Film Uniformity map shows a red warning zone; DEFECT RATE rises.

Instructor action: Have learners readjust gas-flow distribution and chamber pressure themselves to restore uniformity.

◆ 6D FAULT CASE — Expected Pattern: Edge

Gas flow at the chamber periphery often differs from the center due to showerhead and exhaust geometry, so edge dies are the most common location for step-coverage shortfalls.

> **Instructor:** *This is the third module in this program that can share an Edge signature with Oxidation and Etching. What single reading on this station's own dashboard would let you rule Deposition in or out?*

Wrap-Up

> **Instructor:** *Today's Deposition process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 9's integrated exercise.*

Checkpoint Q&A

Q. Why does ALD give better thickness control than CVD?

A. Its self-limiting reaction deposits exactly one atomic layer per cycle, giving high thickness repeatability.

Q. What is step coverage, and why does it matter?

A. How uniformly a film covers the sidewalls and bottom of a pattern — it governs the electrical connection reliability of high-aspect-ratio structures.

II-6. [Session 6] Ion Implantation

Learning Objectives

- Explain the physical/chemical principles of the Ion process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final wafer map — the 6D fault case.

Intro

> **Instructor:** *Today's process gives pure silicon its electrical properties: ion implantation. We cover how dose and energy determine a device's threshold voltage.*

Development — Simulator-Linked Instruction

- Play the sequence of boron (B) and phosphorus (P) ions being accelerated into the wafer on the STEP 06 screen, and graph how dose and energy affect implant depth and concentration.
- Cover post-implant annealing, which repairs lattice damage and activates the dopants.
- Learner exercise: run a simulation mission adjusting dose and energy to hit a target threshold voltage (Vth).

⚠️ PROCESS ANOMALY SCENARIO — Threshold Voltage (Vth) Deviation from Dose Error

Root cause: Ion beam current instability causes the actual implant dose to drift from the setpoint.

Symptom (dashboard reaction): The Vth prediction graph exits the spec range; YIELD TREND declines.

Instructor action: Demonstrate correcting beam current and recalculating implant time to bring dose back to target.

◆ 6D FAULT CASE — Expected Pattern: Donut

Beam-scan zone effects can leave a specific radial band under- or over-dosed while the center and far edge remain within spec, the same mechanism that produces Photolithography's Donut signature.

> **Instructor:** *If beam current drifted only during the scan pass covering one particular radius, what would the resulting wafer map shape tell you — and what would rule out a Photolithography cause instead?*

Wrap-Up

> **Instructor:** *Today's Ion process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 9's integrated exercise.*

Checkpoint Q&A

Q. Why is annealing needed after ion implantation?

A. To repair the crystal structure damaged by ion collisions and electrically activate the implanted dopants.

Q. What do dose and energy each determine?

A. Dose determines dopant concentration (threshold voltage); energy determines implant depth (junction depth).

II-7. [Session 7] Metallization

Learning Objectives

- Explain the physical/chemical principles of the Metallization process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final wafer map — the 6D fault case.

Intro

> **Instructor:** *Today's process electrically connects the finished devices: metallization. We cover reliability issues in copper and aluminum interconnects.*

Development — Simulator-Linked Instruction

- Play the sequence of copper interconnect formation via the Damascene process (trench etch → barrier layer → copper plating → CMP planarization) in order on the STEP 07 screen.
- Explain the relationship between interconnect resistance and current density, introducing the concept of electromigration.
- Learner exercise: narrow the interconnect width step by step in the simulator, observing how resistance and reliability indicators degrade, and derive a design margin.

⚠️ PROCESS ANOMALY SCENARIO — Rising Interconnect Resistance / Open Circuit from Void Formation

Root cause: Poor trench fill during copper plating leaves microscopic voids inside the interconnect.

Symptom (dashboard reaction): Measured interconnect resistance spikes sharply; the 3D view highlights the void inside the interconnect.

Instructor action: Have learners readjust plating current density and barrier-layer thickness themselves to improve fill quality.

◆ 6D FAULT CASE — Expected Pattern: Edge

Plating current density and CMP pad conditioning both have well-documented edge effects, making Metallization the fourth module in this program that can produce an Edge-shaped pattern.

> **Instructor:** *Four different modules in this program — Oxidation, Etching, Deposition, and now Metallization — can all produce the same Edge shape on a wafer map. What does that tell you about how much weight to put on shape alone, versus each station's own dashboard reading?*

Use this as the capstone teaching moment for the Edge family: the shape narrows 13 possible stations down to 4 — a large and genuinely useful reduction — but confirming which of the 4 requires checking each station's own data, exactly the discipline the Session 9 root-cause tracing exercise will test.

Wrap-Up

> **Instructor:** *Today's Metallization process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 9's integrated exercise.*

Checkpoint Q&A

Q. What is the role of CMP (chemical mechanical polishing) in the Damascene process?

A. It removes overburden copper and planarizes the surface to prepare for the next interconnect layer.

Q. What is electromigration?

A. A reliability degradation phenomenon where high current density causes metal atoms to migrate, creating voids or open circuits in the interconnect.

II-8. [Session 8] EDS (Electrical Die Sorting)

Learning Objectives

- Explain the physical/chemical principles of the EDS process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final wafer map — the 6D fault case.

Intro

> **Instructor:** *Today is the final front-end gate: EDS, which tests whether each individual chip on a finished wafer functions correctly. The data from this stage determines the final yield.*

Development — Simulator-Linked Instruction

- Play the sequence of a probe card contacting individual dies on the wafer to measure electrical signals on the STEP 08 screen.
- Explain how the wafer map colors good (Pass) and bad (Fail) dies, and how to interpret spatial fail patterns (edge fail, center fail, etc.) as clues to which process anomaly occurred.
- Learner exercise: given a randomly generated wafer map, complete a diagnostic worksheet estimating which process stage the problem originated in (applying everything learned across Sessions 1–7).

⚠️ PROCESS ANOMALY SCENARIO — False Rejects from Probe Contact Failure

Root cause: Worn or misaligned probe pins raise contact resistance, causing good dies to be misjudged as defective.

Symptom (dashboard reaction): YIELD reads lower than reality, while the underlying DEFECT RATE data is distorted.

Instructor action: Demonstrate recalibrating the probe card to restore measurement reliability. Use this scenario to emphasize the skill of distinguishing 'an error in the measurement itself' from 'an actual process defect.'

◆ 6D FAULT CASE — Expected Pattern: Local Cluster

A single worn or misaligned probe pin affects only the dies it physically contacts — a small, tight group on the map, unrelated to the wafer's own center or edge geometry.

> **Instructor:** *Unlike the four Edge-family modules, this pattern doesn't point back to an earlier process at all. If you see a Local Cluster on the final wafer map, should your first move be to check Oxidation's furnace zones, or something at this very station?*

This is the payoff for Session 0's distinction between in-process and process-result anomalies: a Local Cluster here is a measurement-station problem, not a wafer-fabrication problem — the wafer itself may be perfectly fine.

Wrap-Up

> **Instructor:** *Today's EDS process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 9's integrated exercise.*

Checkpoint Q&A

Q. What does an edge-fail pattern on a wafer map typically indicate?

A. It often points to a process uniformity issue at the wafer edge — such as the edge effect in deposition or etching — related to temperature or gas-flow non-uniformity.

Q. How does EDS connect front-end and back-end?

A. Only dies that pass EDS are carried forward to back-end (packaging); failing dies are filtered out beforehand to avoid wasting back-end effort.

II-9. [Session 9] Integration · 8-Process Chain Scenario & Certification

Learning Objectives

- Operate all eight front-end stages as a single, continuous digital-twin scenario.
- Trace which process stage a specific final-yield anomaly originated from — root-cause tracing.
- Verify learning outcomes through a certification assessment at the end of the program.

Intro

Instructor: *Today we connect everything you learned over the past eight weeks into a single wafer journey. You now have both the expertise of each individual process and the perspective of an engineer surveying the entire line.*

Development — Full-Scenario Practice

- Run the Full Digital Twin Dashboard to operate the integrated simulation where all eight stages, Wafer Fabrication through EDS, are sequentially linked.
- The instructor secretly inserts an anomaly condition into one of the eight processes beforehand. Learners see only the final EDS wafer map's defect pattern and trace back the originating process in teams (40-minute diagnostic mission).
- Each team presents its diagnosis and reasoning; the instructor compares it against the true origin and gives feedback.

Certification Assessment

- Written assessment: 20 questions covering core concepts from every session (25 min), plus a practical assessment: diagnose and respond to one anomaly scenario in the simulator (20 min).
- Passing criteria: 70+ on the written assessment, and 6 of 8 items met on the practical checklist, to receive a certificate of completion.

Wrap-Up

Instructor: *As of today, you can diagnose the entire journey from a single wafer to a finished chip. Well done.*

Checkpoint Q&A

Q. Why does root-cause tracing matter in the 8-process integrated scenario?

A. A final defect can be the accumulated result of several process steps, so surface-level results (EDS data) alone are not enough — the full process history must be traced for accurate root-cause identification.

Q. Why does the certification assessment include a practical component?

A. To verify diagnostic and response capability in an actual anomaly situation, not just rote knowledge.

IEG co., Ltd.

Semiconductor Back-End 6D Digital Twin Simulator

Instructor Script Guide

7 Sessions · 60–120 min each · IEG Training Series

IEG Co., Ltd. — Global Business Division

I. Program Overview

This guide provides session-by-session scripts for teaching the five back-end stages (Die Bonding → Wire Bonding → Molding → Saw/Dicing → Package Testing) using IEG's 6D Digital Twin Simulator.

Defining the '6D Digital Twin'

- 3D — a realistic spatial model of actual equipment, dies, and packages
- 4D — reflects process progression along a time axis (cure cycles, bonding sequence, etc.)
- 5D — a data/intelligence layer where AI predicts and warns of anomalies from accumulated process data (the dashboard's five indicators)
- 6D — a spatial pattern intelligence layer that reads the package map's spatial pattern (Edge · Center · Donut · Scratch · Random · Local Cluster) at Package Testing to narrow down root-cause candidates

This Guide's Approach to Fault Cases

Every module session in this guide pairs its numeric Process Anomaly Scenario with a 6D Fault Case: a prediction of what spatial pattern that specific fault would produce on the final package map, and why. Two modules (Wire Bonding and Saw/Dicing) are deliberately given the same expected pattern — Scratch — for two different physical reasons, so learners practice the key 6D lesson: shape narrows the search, but does not replace checking a station's own data.

Session Overview

Session	Topic	Key Simulator Module	Duration
0	Orientation · 6D Digital Twin Overview (Back-End)	Full Dashboard / 3D View Navigation	60 min
1	Die Bonding	MODULE 01 · Collet Pick & Place / Cure	90 min
2	Wire Bonding	MODULE 02 · EFO · Loop · Vision Alignment	90 min
3	Molding	MODULE 03 · EMC · PID Control · Six Sigma Analysis	90 min
4	Saw / Dicing	MODULE 04 · SFC Design · EtherCAT · Reject Logic	90 min
5	Package Testing	MODULE 05 · AOI · X-ray · Bond Test · SPC	90 min
6	Integration · 5-Process Chain Scenario & Certification	Full Digital Twin Dashboard (fully linked)	120 min

** Each session runs in three stages: a 10-minute intro, 50–90 minutes of development/simulator practice, and 15–20 minutes of wrap-up and checkpoint review.*

II. [Session 0] Orientation · 6D Digital Twin Overview (Back-End)

Learning Objectives

- Understand the full flow of the five back-end modules (Die Bonding → Wire Bonding → Molding → Saw/Dicing → Package Testing).
- Explain how back-end's core purpose (mechanical assembly and reliability) differs from front-end.
- Understand the 6D digital twin concept (3D geometry + 4D time + 5D data/AI + 6D spatial pattern intelligence) in a back-end context.

Intro

> **Instructor:** *Once a wafer finishes front-end testing, its individual dies are cut apart and carried through the five processes you'll learn starting today, ending as the chip inside your phone or your car.*

> **Instructor:** *If front-end was a game of precisely controlling chemical and physical reactions, back-end is a game of mechanical assembly precision and reliability. We attach the die (Die Bonding), wire it up (Wire Bonding), encapsulate it (Molding), cut it apart (Dicing), and inspect it (Testing) — five stages.*

Development — Mapping the Five Back-End Processes

On the whiteboard, list the five processes in order and have learners summarize in one line what physically happens to the package at each stage.

Reviewing the 6D Digital Twin Definition

- 3D — realistic physical geometry of the equipment, dies, and packages
- 4D — process progression over time (cure cycles, bonding sequence, etc.)
- 5D — a data/AI layer that predicts and flags anomalies from the dashboard's five numeric indicators
- 6D — spatial pattern intelligence: reading the shape of failures on a package map at Package Testing (Edge · Center · Donut · Scratch · Random · Local Cluster) to narrow down root-cause candidates

Wrap-Up

> **Instructor:** *The five-process map you drew today is your navigation for the next five weeks. Starting next session, we'll operate Die Bonding hands-on.*

Checkpoint Q&A

Q. How does back-end's core purpose differ from front-end?

> A. Back-end physically assembles and packages the finished dies from a wafer into a mountable product, so mechanical reliability and assembly precision are the central concerns.

Q. Why does 6D matter especially in back-end?

> A. The spatial pattern that appears on the final Package Testing map is a key clue for tracing back which of the five processes the problem actually started in.

II-1. [Session 1] Die Bonding

Learning Objectives

- Explain the principles and reliability-management points of the Die process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

> **Instructor:** *Today is the first back-end stage: attaching an individual die to a leadframe or substrate with epoxy. Collet pick-and-place precision and cure management are central.*

Development — Simulator-Linked Instruction

- On the MODULE 01 screen, play the sequence where the collet vacuum-picks a die from the wafer and places it precisely onto the epoxy-dispensed leadframe.
- Explain how epoxy dispense volume and pick-and-place alignment error affect final reliability — under-dispense leaves voids beneath the die, and misalignment leads to downstream wire-bonding failures.
- Learner activity: adjust the Bond Force and cure-temperature sliders and observe how Defect Rate changes, recording results in the worksheet.

⚠️ PROCESS ANOMALY SCENARIO — Void Formation from Under-Dispense

Root cause: A partially clogged dispenser nozzle drops the dispense volume below the 100% spec, leaving a void beneath the die.

Symptom (dashboard reaction): Defect Rate spikes sharply; Alignment Error also drifts out of spec at the same time.

Instructor action: Have learners readjust the dispense-volume and alignment sliders and demonstrate the recovery to normal range.

◆ 6D FAULT CASE — Expected Pattern: Local Cluster

A clogged dispenser nozzle degrades gradually, so a run of consecutively bonded units — not the whole tray — shares the defect before the drift is caught.

> **Instructor:** *If this nozzle clog affected roughly 20 consecutive units before an operator caught it, what would you expect the final package map to look like? Not the whole tray — just a tight group.*

Contrast this with a fully random single-die defect: a clog-driven fault clusters in production sequence, while an isolated particle contamination event does not.

Wrap-Up

> **Instructor:** *Today's Die process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. What is the difference between DAF and liquid epoxy?

A. DAF (Die Attach Film) offers highly uniform thickness; liquid epoxy is flexible but requires tighter dispense-volume control.

Q. What is the bottleneck of the Die Bonding process?

A. The cure oven's batch cycle time, which is difficult to parallelize and limits overall line throughput.

II-2. [Session 2] Wire Bonding

Learning Objectives

- Explain the principles and reliability-management points of the Wire process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

> **Instructor:** *Today's process electrically connects die pads to the leadframe using fine gold or copper wire. EFO ball formation and capillary loop precision are the focus.*

Development — Simulator-Linked Instruction

- On the MODULE 02 screen, play the sequence where an EFO spark forms a ball at the wire tip, and the capillary traces an arched loop to the leadframe.
- Explain that too-low a loop height risks wire sag and shorting with neighboring wires, while too-high a loop risks wire sweep during molding.
- Learner activity: adjust EFO current and loop-height sliders and observe how ball size and loop shape change, comparing against the literature-validated optimal EFO current (~44 mA).

⚠️ PROCESS ANOMALY SCENARIO — Loop Height Deviation Causing Wire Sag / Heel Crack

Root cause: Z-axis servo drift on the capillary pushes loop height outside the 150±15 µm spec, causing wire sag or heel cracking.

Symptom (dashboard reaction): Defect Rate rises; EFO current also drifts outside the 44±8 mA spec at the same time.

Instructor action: Have learners readjust the loop-height and EFO-current sliders and demonstrate the recovery to normal range.

◆ 6D FAULT CASE — Expected Pattern: Scratch (Line)

Multi-head wire bonders assign fixed tray positions to each bonding head. If one specific head drifts out of spec while the others remain fine, every unit that head touched forms a straight line or row on the map — not because of physical contact damage, but because of a fixed equipment-to-position mapping.

> **Instructor:** *This is one of the more advanced fault cases: a Scratch-shaped pattern normally means physical handling damage. But here, the exact same shape can come from a single bonding head being out of spec. How would you tell the two apart using only this station's dashboard, without opening the tool?*

Key teaching point: the same spatial shape can come from two very different root causes (mechanical damage vs. one bad equipment head). The shape narrows your search — it does not replace checking this station's own parameter readings.

Wrap-Up

> **Instructor:** *Today's Wire process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. What is the difference between a ball bond and a wedge bond?

A. A ball bond forms a ball via EFO before thermocompression (typically the 1st bond); a wedge bond compresses directly without a ball.

Q. Why does Loop Height matter for downstream reliability?

A. Too low risks shorting from wire sag; too high risks wire sweep during molding.

II-3. [Session 3] Molding

Learning Objectives

- Explain the principles and reliability-management points of the Molding process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

> **Instructor:** *Today's process encapsulates the die and wires in Epoxy Molding Compound (EMC). Mold-fill control across the preheat–cure–cool profile is central.*

Development — Simulator-Linked Instruction

- On the MODULE 03 screen, play the sequence where the mold clamps and EMC resin fills the cavity under transfer pressure.
- Explain the three-stage temperature profile (preheat → cure → cool), and how PID overshoot causes premature EMC curing and incomplete fill.
- Learner activity: adjust transfer-pressure and cure-time sliders and observe how Defect Rate responds when values fall outside the target windows (pressure 70±8 kgf/cm², cure time 90±15 sec).

⚠️ PROCESS ANOMALY SCENARIO — Low-Pressure Fill Causing Incomplete Runner Fill / Void

Root cause: Transfer pressure drops below spec, so the mold cavity does not fully fill with EMC and voids form.

Symptom (dashboard reaction): Defect Rate rises; Yield Trend shifts to a downward slope.

Instructor action: Have learners readjust the transfer-pressure parameter and demonstrate the fill-quality recovery.

◆ 6D FAULT CASE — Expected Pattern: Center

Multi-cavity molds are standard in production. Center cavities of a mold platen tend to run hotter than edge cavities due to heat concentration, so if the cure profile drifts, center-cavity units cure differently from edge-cavity units.

> **Instructor:** *If your mold has 16 cavities arranged in a grid and the center cavities are consistently a few degrees hotter, what shape would you expect on the map once those units are laid out in their original tray positions?*

This is the back-end mirror of front-end's Center pattern (chuck hot-spot) — same shape, same underlying idea (a hot zone in the middle of a fixture), completely different physical stage of the flow.

Wrap-Up

> **Instructor:** *Today's Molding process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. Why is minimizing PID overshoot important?

A. If cure temperature exceeds target, the EMC cures prematurely, causing fill defects and elevated internal stress.

Q. What is the primary cause of package warpage?

A. CTE (coefficient of thermal expansion) mismatch between the EMC and the leadframe/substrate causes uneven shrinkage on cooling.

II-4. [Session 4] Saw / Dicing

Learning Objectives

- Explain the principles and reliability-management points of the Saw process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

> **Instructor:** *Today's process cuts the molded strip into individual packages with a diamond blade. Blade-wear management and SFC (equipment controller) design are central.*

Development — Simulator-Linked Instruction

- On the MODULE 04 screen, play the sequence of the high-speed rotating blade cutting the strip while coolant sprays.
- Explain that as a blade wears, cutting resistance rises, rotation speed becomes unstable, and micro-chipping appears at die edges.
- Learner activity: adjust blade-rotation-speed and coolant-flow sliders and observe how chipping rate (Defect Rate) responds. Introduce how SFC (Smart Factory Controller) design and EtherCAT communication let a real line monitor blade condition in real time.

⚠️ PROCESS ANOMALY SCENARIO — Blade Wear Causing Chipping Defects

Root cause: Progressive blade wear raises cutting resistance, pushing rotation speed out of spec and causing micro-chipping at die edges.

Symptom (dashboard reaction): Defect Rate rises; coolant flow also drops below spec at the same time.

Instructor action: Have learners readjust rotation-speed and coolant-flow parameters and demonstrate the recovery to normal range.

◆ 6D FAULT CASE — Expected Pattern: Scratch (Line)

A dicing saw cuts along fixed streets (lanes) across the strip. If one specific street develops a problem — a locally worn section of blade edge, a resin buildup, or a kerf-width deviation on that lane — every package along that one street shows chipping, forming a literal straight line.

> **Instructor:** *Notice this is the second module in a row that can produce a Scratch pattern — but for a completely different physical reason than Wire Bonding's. Here, the line runs along a physical cutting path; at Wire Bonding, it followed an equipment-head assignment. How does knowing which module you're looking at change how you'd confirm the cause?*

Use this back-to-back Scratch example deliberately: it proves the point from Session 2 that identical map shapes require checking the originating station's own dashboard data to fully confirm the mechanism, not just the shape.

Wrap-Up

> **Instructor:** *Today's Saw process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. What is the difference between blade dicing and stealth dicing?

A. Blade dicing cuts physically; stealth dicing forms an internal modified layer with a laser, then separates mechanically with less chipping.

Q. What happens when coolant flow is insufficient?

A. Heat builds up on the blade, accelerating wear and increasing both chipping rate and thermal damage.

II-5. [Session 5] Package Testing

Learning Objectives

- Explain the principles and reliability-management points of the Package process.
- Use the simulator to understand the cause-and-effect relationship between key process parameters and result indicators.
- Diagnose and respond to a representative anomaly scenario for this process.
- Predict the spatial pattern a fault at this station would produce on the final package map — the 6D fault case.

Intro

> **Instructor:** *Today is the final back-end gate: inspecting the finished package's appearance and electrical characteristics. AOI, X-ray, Pull Test, and 6D spatial pattern reading are all central here.*

Development — Simulator-Linked Instruction

- On the MODULE 05 screen, play the sequence of a socket contacting an individual package to measure electrical signals while an AOI camera scans.
- Explain how the AOI decision threshold is set and periodically recalibrated, and how a threshold set too loosely lets genuinely defective units pass as good (a false accept).
- Learner activity: use the new Package Map Pattern Drill panel to practice reading all six spatial patterns (Edge · Center · Donut · Scratch · Random · Local Cluster) hands-on — use alongside the companion Wafer Map Pattern Recognition Reference Card and Scripted Guide.

⚠️ PROCESS ANOMALY SCENARIO — AOI Threshold Drift Causing False Accepts

Root cause: Illumination degradation shifts the AOI decision threshold outside the 50±8 spec, causing genuinely defective packages to pass as good.

Symptom (dashboard reaction): Yield reads higher than reality, while the underlying Defect Rate data is quietly distorted.

Instructor action: Demonstrate readjusting the AOI-threshold parameter to restore inspection reliability. Use this scenario to emphasize the difference between 'an error in the inspection itself' and 'an actual process defect.'

◆ 6D FAULT CASE — Expected Pattern: Random (Scatter)

An AOI threshold drift is a calibration problem, not a location-based defect — it doesn't care where on the tray a unit sits. Any unit, regardless of position, has an equal chance of being misjudged, so the map shows no shape at all.

> **Instructor:** *This is the most important fault case in the whole back-end series, because it's the one where 6D correctly tells you there IS no spatial story. If a team sees a random scatter and starts hunting for a station with a location-based cause, what should you tell them?*

This deliberately completes the set: four of the five back-end modules can produce a shape (Cluster, two different Scratch causes, Center); this one produces no shape at all — and recognizing 'no shape' as diagnostic information is itself an advanced 6D skill, not a failure to find one.

Wrap-Up

> **Instructor:** *Today's Package process leads directly into next session's process. Keep today's worksheet — you'll use it again in Session 6's integrated exercise.*

Checkpoint Q&A

Q. Why must the false accept rate stay low?

A. A defective unit shipped as good leads to field failure (RMA), which carries far higher cost and reliability risk than catching it at test.

Q. Why does the 6D reading at Package Testing matter?

A. A number alone (Defect Rate) can't reveal the cause, but the spatial shape of a package map (e.g., a straight-line scratch) immediately narrows the candidate causes — including, as this session shows, telling you when there is no spatial cause at all.

II-6. [Session 6] Integration · 5-Process Chain Scenario & Certification

Learning Objectives

- Operate all five back-end modules as a single, continuous digital-twin scenario.
- Trace a final Package Testing result back to the originating process stage — root-cause tracing.
- Verify learning outcomes through a certification assessment that includes 6D spatial pattern reading.
- Use the 6D Fault Case Gallery to recognize that identical shapes can come from different stations and different mechanisms.

Intro

> **Instructor:** *Today we connect everything you learned over the past five weeks into a single package journey. You now have both the expertise of each individual process and the perspective of a reliability engineer surveying the entire line.*

Development — Full-Scenario Practice

- Run the Full Digital Twin Dashboard to operate the integrated simulation where all five stages, Die Bonding through Package Testing, are sequentially linked.
- The instructor secretly inserts an anomaly condition into one of the five processes beforehand. Learners see only the final Package Testing result — both the numeric data and the spatial pattern — and trace back the originating process in teams (40-minute diagnostic mission).
- Each team presents its diagnosis and evidence (numeric data + pattern shape); the instructor compares it against the true origin and gives feedback.

6D Fault Case Gallery — Reference

Review this table with learners before the diagnostic mission. It summarizes every fault case covered this program, including the deliberate overlap between Wire Bonding and Saw/Dicing.

Module	Fault	Expected 6D Pattern	Why
Die Bonding	Under-dispense void	Local Cluster	Nozzle clog degrades gradually — a run of consecutive units is affected, not the whole tray.
Wire Bonding	Loop height / EFO drift	Scratch (Line)	One bonding head, fixed to specific tray positions, drifts out of spec while others stay fine.
Molding	Low-pressure incomplete fill	Center	Center cavities of a multi-cavity mold run hotter, curing differently from edge cavities.
Saw / Dicing	Blade wear chipping	Scratch (Line)	One dicing street/lane develops a local blade issue — every unit along that lane is affected.
Package Testing	AOI threshold drift	Random (no shape)	A calibration problem has no location dependence — any unit can be equally affected.

Discussion: two rows share the same 'Scratch' pattern. Ask learners how they would distinguish a Wire Bonding cause from a Saw/Dicing cause using only the shape — the honest answer is that they can't, until they check each station's own dashboard readings.

Certification Assessment

- Written assessment: 15 questions covering core concepts from every session plus 6D pattern reading (20 min).
- Practical assessment: diagnose and respond to one anomaly scenario in the simulator, plus read one spatial pattern correctly (25 min).

- Passing criteria: 70+ on the written assessment, and 5 of 6 items met on the practical checklist, to receive a certificate of completion.

Wrap-Up

Instructor: *As of today, you can diagnose the entire journey from a single die to a finished package. Combined with the front-end program, you're now ready to take on the full 13-station Wafer to Package Capstone. Well done.*

Checkpoint Q&A

Q. Why does root-cause tracing matter across a 5-process chain?

A. A final defect can result from accumulated deviations across multiple steps, so the full process history — and the spatial evidence — must be examined rather than just the final test result.

Q. Why include 6D pattern reading in the certification assessment?

A. Number-based diagnosis and spatial-pattern-based diagnosis require different reasoning processes, and real yield-review meetings on the shop floor require both abilities together.